

MATA37H3 — Week 13

Answer Key & Misconception Notes

For tutor use only — do not show the student until after their attempt. (Taylor & Maclaurin Series: Analytic Functions)

EXAMPLE 1 — Answer

Every derivative of e^x is e^x itself, and on $[-R, R]$, $e^t \leq e^R$ for all t in that interval — so $M = e^R$ bounds $|f^{(n+1)}(t)|$ uniformly in both $t \in [-R, R]$ and n . Then $|R_n(x)| \leq \frac{e^R}{(n+1)!} |x|^{n+1} \rightarrow 0$ as $n \rightarrow \infty$ for any fixed $x \in [-R, R]$ (factorial beats any fixed base).

Common misconception: Trying to use $M = e^x$ itself as the bound (a variable, not a constant) — the bound **must** be a single number that works for the whole interval, not a function of t .

EXAMPLE 2 — Answer

The bound $M = e^R$ depends on R — it isn't a single constant working for *all* of $\mathbb{R}$ simultaneously. Instead: fix any $x \in \mathbb{R}$, choose $R = |x|$ (or larger), apply Problem 1's argument on $[-R, R]$, and conclude $R_n(x) \rightarrow 0$ for *that* x — repeating this for every x shows e^x is analytic on all of $\mathbb{R}$, just interval-by-interval rather than with one universal M .

Common misconception: Assuming “analytic on every bounded interval” is somehow weaker than “analytic on $\mathbb{R}$ ” — together, “analytic on $[-R, R]$ for every R ” **is** exactly what “analytic on $\mathbb{R}$ ” means, since every real number lies in some $[-R, R]$.

EXAMPLE 3 — Answer

$\frac{1}{1-x^3} = \sum_{n=0}^{\infty} x^{3n} = 1 + x^3 + x^6 + x^9 + \dots$ (substitute x^3 into the geometric series, valid $|x| < 1$). Coefficient of x^{15} : since $15 = 3n \Rightarrow n = 5$, the coefficient is 1. Matching with $\frac{f^{(15)}(0)}{15!} = 1 \Rightarrow f^{(15)}(0) = 15!$.

Common misconception: Forgetting that most coefficients in this series are 0 (only multiples of 3 appear) — if asked for, say, $f^{(16)}(0)$ instead, the answer would be 0, not “undefined.”

EXAMPLE 4 — Answer

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$; at $x = 1$: $e^1 = \sum_{n=0}^{\infty} \frac{1}{n!} = A$. $A = e$.

Common misconception: Trying to compute this as an unfamiliar numerical series rather than instantly recognizing the pattern $\frac{1}{n!}$ as the e^x series evaluated at $x = 1$ — this recognition should be immediate once the six core series are memorized.

EXAMPLE 5 — Answer

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}; \text{ at } x = \frac{\pi}{4}: \sin \frac{\pi}{4} = \sum_{n=0}^{\infty} (-1)^n \frac{(\pi/4)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n+1}}{4^{2n+1} (2n+1)!} =$$

B . So $B = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$.

Common misconception: Confusing the $\sin x$ series (odd powers over $(2n+1)!$) with the $\arctan x$ series (odd powers over $2n+1$, no factorial) — the presence of $(2n+1)!$ in the denominator is the signature of $\sin x$, not $\arctan x$.

EXAMPLE 6 — Answer

$$e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}, \text{ valid for all } x \text{ (substitution never shrinks an infinite radius).}$$

Common misconception: Writing $e^{-x} = -e^x$ or otherwise trying to relate it algebraically to e^x as a whole, instead of substituting directly into the series (they are **not** negatives of each other).

EXAMPLE 7 — Answer

$$\frac{e^x + e^{-x}}{2} = \frac{1}{2} \left[\sum_{n=0}^{\infty} \frac{x^n}{n!} + \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!} \right] = \sum_{n=0}^{\infty} \frac{1 + (-1)^n}{2} \cdot \frac{x^n}{n!} = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} \text{ (all odd-} n \text{ terms}$$

cancel since $1 + (-1)^n = 0$ there; even- n terms double then get halved back). This makes sense because f is even ($f(-x) = f(x)$), and an even function's Maclaurin series can only contain even powers of x — odd-power coefficients must vanish identically.

Common misconception: Not connecting the algebraic cancellation to the general fact about even/odd functions — this is a reusable shortcut: even $f \Rightarrow$ series has only even powers; odd $f \Rightarrow$ only odd powers; check this *before* grinding through the addition.

EXAMPLE 8 — Answer

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}; \text{ and } \ln(1-x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(-x)^n}{n} = \sum_{n=1}^{\infty} (-1)^{n+1} (-1)^n \frac{x^n}{n} = -\sum_{n=1}^{\infty} \frac{x^n}{n}. \text{ So } \ln \frac{1+x}{1-x} = \ln(1+x) - \ln(1-x) = \sum_{n=1}^{\infty} [(-1)^{n+1} + 1] \frac{x^n}{n} = \sum_{k=0}^{\infty} \frac{2}{2k+1} x^{2k+1}$$

(only odd n survive), valid for $|x| < 1$.

Common misconception: Substituting $-x$ for x in the $\ln(1+x)$ series but forgetting that $(-1)^{n+1}(-x)^n$ simplifies to a **constant sign** (-1 , not alternating) once you track $(-1)^n$ separately.